PSYC4411

CHALLENGE LAB: PREDICT & EXPLAIN REGRESSION ON REAL DASS DATA

Week 4 · Semester 1, 2026

Current Advances in Psychological Methods & Analyses

TODAY'S CHALLENGE

- Build **regression models** to predict DASS Depression scores from personality and demographics
  - Baseline → Linear Regression → Ridge → Lasso
- Report **cross-validated MAE and R^2** — no peeking at the test set until the end
  - Remember from Week 3: the test set stays locked until your final evaluation
- **Interpret the coefficients** — which personality traits predict depression? Does the model match psychological theory?
- Prepare **1 HTML slide** (~3 min) for Week 5

THE DATASET

OPENPSYCHOMETRICS DASS-42

- **39,775** real respondents from around the world
- **172 columns:** DASS-42 items, Big Five personality (TIPI), demographics, response times, vocabulary check
- **Target:** DASS Depression subscale (14 items, scored 0–42)
- **Features:** Big Five personality traits + demographics (age, gender, education, ...)

DATA QUALITY CHALLENGES

- Some ages are clearly wrong (e.g., 1998 — a birth year, not an age)
- ~76% female, ~54% from Malaysia — not a representative sample
- Vocabulary check with fake words to catch careless responders
- Tab-separated file — use `sep="\t"` when loading

NEW SKILL: DEBUGGING WITH AI

Week 2 was about **prompting**. This week is about **debugging** — when code breaks, how to get useful help from your AI assistant.

Weak:
"My code doesn't work. Fix it."

Strong:
"I'm getting `ValueError: could not convert string to float: 'Female'` when fitting Ridge regression. My features include Gender (coded 1/2/3). I expected numeric values but something went wrong. What caused this and how do I fix it?"

- **Share the full error** — copy the entire traceback, not just the last line
- **Describe expected vs actual** — "I expected X but got Y"
- **Ask for explanation** — "Why did this happen?" not just "Fix this"

🚀 GETTING STARTED

OPEN YOUR STARTER FILE

1. `starter.ipynb` (notebook) or `starter.py` (script)
2. Run the setup cells — imports, data loading, basic cleaning are done for you
3. **Ask the AI to plan first** — don't jump straight to modelling

YOUR JOB

4. Score the DASS & TIPI (with AI help)
5. Build & cross-validate models
6. Interpret coefficients
7. Final test set evaluation

KEY NUMBERS TO AIM FOR

Baseline MAE: ~10.6

Good model MAE: ~8.2

Good model R^2: ~0.34

R^2 of 0.34 is strong for behavioural science — personality explains about a third of the variance in depression scores

Full details in the challenge brief:
`weeks/week-04-lab/README.md`